STiMUS-HAI (v3.1) ODD Protocol Document

Model: STiMUS-HAI — Stigmergic-Mutualistic IMOI Model, Human-AI extension

Platform: NetLogo 7

Version: 3.1

Date: 2026-07-13

Author: Yevgeny Patarakin (HSE University / Moscow City University)

This document follows the ODD (Overview, Design concepts, Details) protocol for describing individual- and agent-based models (Grimm et al. 2006, 2010). Sections 1–7 are the core ODD protocol, extending the v2.1 ODD with the human-AI mechanics introduced in v3.0/v3.1. Section 8 onwards document simulation experiments, interpretation, and limitations and are not part of the core ODD.

1. Purpose

1.1 Overall purpose

STiMUS-HAI simulates the emergence of collaborative-editing patterns in a socio-technical system where human and AI contributors jointly enrich a shared repository of digital artefacts (e.g., wiki pages, shared documents, open educational resources, code files). It extends the base STiMUS model (v2.2, human-only) with two additional theoretical questions:

1. **Does mixing AI agents into a human team distort measures of humancoordination in ways that aggregate indicators fail to show?** (v3.0 question, tested by experiment E11.)
2. **Does AI's effect on team outcomes depend on the *type of work AI isasked to do — routine/predictable versus judgment-requiring — not just onhow much AI is present?*** (v3.1 question, tested by experiment E12.)

The original v2.1 theoretical claim — that stigmergy and mutualism are orthogonal, independently designable coordination levers — is preserved unchanged as the model's foundation (§1 of the v2.1 ODD, hypotheses H1–H5, still holds at ai-share = 0).

1.2 What v3.0/v3.1 add

- **v3.0** introduces a second breed, `ai-agents`, alongside `humans`. Both breeds follow identical target-selection, pheromone, and mutualism rules — deliberately, so that any behavioural difference is attributable to team *composition*, not a built-in advantage of one breed. The one designed

asymmetry: AI agents never accumulate `shared-mental-model`, and their `motivation` is a fixed constant rather than an updated state variable.

- **v3.1** adds a `task-type` dimension to artefacts (`"prediction"` vs. `"judgment"`, controlled by a new `judgment-share` slider) and an `ai-trust` mechanic for humans: trust in AI contributions is built or eroded based on the observed quality of AI-authored work, and gates how much mutualistic benefit a human derives from continuing an AI's work.
- Both extensions collapse to a no-op at their neutral default: `judgment-share = 0` reproduces v3.0 exactly; `judgment-share = 0` **and** `ai-share = 0` together reproduce v2.2 exactly (see §5 and the E11/E12 backward-compatibility checks in §8).

1.3 Hypotheses addressed

Carried over from v2.1 (still hold at `ai-share = 0`): H1 (decoupling), H2 (visibility as master variable), H3 (topic structure as a lever), H4 (tight explicit-assignment filter), H5 (mutualism sustains engagement, not peak quality). See the v2.1 ODD for full statements; not re-derived here.

New for v3.0/v3.1:

- **H-E11.1 / H1 (moderate AI does not disrupt human cognition)** — tested by E11. **Not fully supported**: even a modest AI share (10%) measurably lowers `human-smm`.
- **H-E11.2 / H2 (AI offloads coordination burden without harming outcomes)** — tested by E11. **Partially supported**: aggregate productivity indicators (quality, task-completion-rate, coediting-density) are essentially unaffected by `ai-share` (all $\eta^2 < 0.003$).
- **H-E11.3 / H3 (aggregate ESR masks a real human-human coordination drop)** — tested by E11. **Strongly supported**: `ESR-all` is nearly flat across `ai-share` ($\eta^2 \approx 0.000009$) while `ESR-HH` falls sharply ($\eta^2 = 0.79$).
- **H-E12.1 (judgment-share steepens AI's dilution effect on ESR-HH)** — tested by E12. **Rejected**: the `ai-share` × `judgment-share` interaction on `ESR-HH` is negligible ($\eta^2 \approx 0.00015$).

- **H-E12.2 (judgment-share accelerates AI-trust erosion)** — tested by E12. **Rejected in its direct form**: the ai-share × judgment-share interaction on ai_trust is small ($\eta^2 = 0.035$) and is better explained as a measurement artifact mediated by quality ($\eta^2 = 0.264$ for quality on ai_trust's relationship) than as a direct causal amplification.
- **H-E12.3 (AI's effect on quality is regime-conditional: judgment-heavy vs.prediction-heavy task mixes)** — tested by E12. **Strongly supported**: in a judgment-heavy regime, quality falls 40% (0.194 → 0.116) as ai-share rises from 0 to 0.9; in a prediction-heavy regime it falls only 8% (0.193 → 0.178) over the same range.

2. Entities, State Variables, and Scales

2.1 Entities

Entity	Breed	Count (default)	Role
Human contributor	humans	num-contributors × (1 − ai-share)	Human editor; mobile agent
AI contributor	ai-agents	num-contributors × ai-share	AI editor; mobile agent; identical movement/edit rules to humans except as noted
Artefact patch	patches	1089 (33×33 world)	A unit of shared content (a "page")
Co-editing link	coedits (undirected)	dynamic	Edge between two contributors who co-edited a patch

2.2 State variables — patches (artefacts)

All v2.1 patch variables (pheromone, quality, completeness, recentness, last-editor-id, edit-count, edit-authors, is-task?, task-complexity, resource-value, reuse-count, maintenance-need, topic-tag) are unchanged. New in v3.0/v3.1:

Variable	Type	Range	Meaning

last-editor-ai?	bool	—	Was the patch's most recent editor an AI agent? (v3.0)
ever-ai-touched?	bool	—	Has this patch ever been edited by an AI agent? (v3.1; used to compute the human-only quality baseline)
task-type	string	"prediction" | "judgment"	v3.1 — flagged at setup per judgment-share
judgment-weight	float	0, or [0.4, 1.0]	v3.1 — 0 for prediction patches; random draw for judgment patches; scales down AI edit-power

2.3 State variables — contributors

All v2.1 contributor variables (skill, motivation, shared-mental-model, specialty, benefit-gain, explicit-mode?) are unchanged in meaning.
New in v3.0/v3.1:

Variable	Type	Range	Meaning
is-ai?	bool	—	v3.0 — breed flag, true for ai-agents
ai-trust	float	[0, 1]	v3.1 — meaningful for humans only; humans start at 0.5 (agnostic prior); AI agents hold an unused placeholder (1.0)
ai-consecutive-successes	int	≥ 0	v3.1 — humans only; counts consecutive non-error observations of AI work; drives trust recovery
ai-error-counter	int	≥ 0	v3.1 — humans only, diagnostic; counts observed AI "error" events

For AI agents specifically: motivation is set once to a fixed constant (ai-motivation-constant = 1.0) and never updated; shared-mental-model is set to 0 and never updated — the model's one designed breed asymmetry.

2.4 Global variables

All v2.1 globals (total-edits, cross-author-edits, ESR, mean-quality, mean-resource-value, mutualism-index, coediting-density, active-pages-share, edge-log, task/explicit-assignment counters) are retained. New v3.0/v3.1 counters and reporters:
human-edits-count, ai-edits-count, hh-cross-author-edits, ai-pheromone-deposits, judgment-edits-count, prediction-edits-count, and reporters ESR-all, ESR-HH, ai-pheromone-share, human-edit-share, ai-edit-share, mean-ai-trust, judgment-edit-share,

prediction-edit-share, human-smm-judgment, human-smm-prediction, mean-ai-error-observations.

2.5 Scales

Unchanged from v2.1: a 33×33 torus of patches; one tick = one collaborative session; default run length 500 ticks.

2.6 Interface sliders

The 11 v1/v2.1 sliders (num-contributors, num-tasks, topic-count, vision-radius, evaporation-rate, diffusion-rate, edit-increment, p-explicit, lambda-smm, mutualism-gain-factor, activity-probability) are unchanged. Two new sliders:

Slider	Default	Range	Role
ai-share	0	[0, 1]	v3.0 — fraction of num-contributors that are ai-agents
judgment-share	0	[0, 1], increment 0.05	v3.1 — fraction of patches flagged task-type = "judgment"

3. Process Overview and Scheduling

3.1 Tick structure

Unchanged in outline from v2.1 (§3.1): (1) every contributor of either breed, subject to activity-probability, runs step-contributor — choose target, move, edit; (2) pheromone diffuses and evaporates; (3) artefact ageing; (4) update-metrics; (5) tick. The agents turtle-set (humans and ai-agents combined) is rebuilt once at setup and iterated identically regardless of breed.

3.2 Target choice

Unchanged from v2.1: the same explicit/stigmergic branching (§3.2 of the v2.1 ODD) applies identically to both breeds. judgment-share and task-type do **not** affect target selection — only editing effectiveness

(§7 below) — preserving the v2.1 decoupling principle that stigmergy governs *where*, not *how effectively*, work happens.

3.3 New subprocesses (v3.0/v3.1)

- **Breed-conditional edit power** (§7.3): AI agents' edit-power is scaled down on judgment-type patches only.
- **AI-trust update** (§7.4, new): fires only for a human editing a patch whose previous editor was an AI agent.
- **Trust-modified benefit** (§7.5, new): modifies the local-benefit formula for that same case.

4. Design Concepts

4.1 Emergence

Unchanged in principle from v2.1: all team-level patterns (ESR split by author-pair type, quality, trust trajectories) emerge from local choices; no contributor — human or AI — has a global view or collective objective.

4.2 Adaptation

Humans adapt as in v2.1 (target and effort). AI agents adapt their *target* identically to humans (same stigmergic/explicit rule) but their *effort* parameters (`motivation`) do not adapt — a deliberate simplification standing in for current AI systems' lack of an intrinsic, feedback-driven motivation analogue. Humans additionally adapt their **trust in AI** (`ai-trust`) based on observed AI work quality — this is the one genuinely new adaptive process in v3.1.

4.3 Fitness / Objectives

Unchanged from v2.1: no explicit fitness function for either breed.

4.4 Sensing

As in v2.1, contributors of either breed sense patch state within vision-radius. Humans additionally "sense" — in an aggregate, not per-observation, sense — the population's current quality distribution via the percentile-based error-detection rule (§7.4): rather than comparing an AI edit to a fixed absolute standard, a human's judgment of whether an AI's prior work was substandard is relative to what is currently typical in the system.

4.5 Interaction

Unchanged mechanisms from v2.1 (indirect stigmergic, indirect mutualistic, direct co-editing network) apply identically to human-human, human-AI, and AI-AI pairs, **except**: (a) AI agents do not contribute to shared-mental-model growth on any succession; (b) a human following an AI's edit undergoes the trust update and trust-modified benefit described in §7.4–7.5.

4.6 Stochasticity

Unchanged in form from v2.1: uniform random initial states; deterministic max-one-of target selection within each mode; activity-probability and p-explicit branches consume random numbers. v3.1 adds one more stochastic draw at setup: a judgment-share fraction of patches receives a random judgment-weight ~ U(0.4, 1.0). 50 repetitions per condition (E11) or 100 (E12 main) are used to estimate central tendencies.

4.7 Collectives

Unchanged from v2.1: emergent co-editing clusters and topic-specialised contributors. v3.0/v3.1 add an emergent human-AI succession structure, measurable via ai-pheromone-share and the ESR-all/ESR-HH gap.

4.8 Observation

All v2.1 monitors are retained. New v3.0/v3.1 reporters (see §2.4) allow observation of: the human-only view of coordination (ESR-HH vs. ESR-all), team composition effects (human-edit-share, ai-edit-share, ai-pheromone-share), trust dynamics (mean-ai-trust, mean-ai-error-observations), and task-type-stratified views

(judgment-edit-share, prediction-edit-share, human-smm-judgment, human-smm-prediction).

5. Initialization

setup performs the v2.1 initialization (patches, tasks, contributors) unchanged, then applies two additional passes:

Task-type flagging (v3.1). After the is-task? pass, every patch is first set to task-type = "prediction", judgment-weight = 0; then round (judgment-share × count patches) patches, chosen at random, are re-flagged task-type = "judgment" with judgment-weight ~ U(judgment-weight-min, judgment-weight-max) = U(0.4, 1.0). At judgment-share = 0, n-of 0 patches selects nothing — every patch keeps the prediction/0 default, restoring v3.0 exactly.

Contributor breed split (v3.0). n-ai-agents = round(num-contributors × ai-share); the remainder are humans. Humans are initialised exactly as v2.2 contributors were (skill ~ U(0.3, 1.0), motivation ~ U(0.4, 1.0), shared-mental-model ~ U(0.1, 0.2), plus v3.1 additions ai-trust = ai-trust-initial = 0.5, ai-consecutive-success = 0, ai-error-counter = 0). AI agents draw skill from the *same* distribution as humans (deliberate neutrality), but motivation = ai-motivation-constant = 1.0 (fixed) and shared-mental-model = 0 (fixed); their ai-trust slot holds an unused placeholder (1.0).

At ai-share = 0, n-ai-agents = 0 and the human-creation block is textually identical to v2.2's contributor creation — same random draws in the same order — so v3.0/v3.1 reproduce v2.2 runs seed-for-seed.

Default run: 500 ticks; 50 repetitions per condition (E11) or 100 (E12 main).

6. Input Data

Unchanged from v2.1: the model uses no external input data. All dynamics are endogenous, driven by the 13 interface sliders (11 original + ai-share + judgment-share) and the fixed internal parameters (§7.6).

7. Submodels

7.1–7.2 Stigmergic target selection and explicit assignment

Unchanged from v2.1 (§7.1–7.2 of the v2.1 ODD); applied identically to both breeds, with resource-value and judgment-weight both absent from the target weight — target choice remains blind to both mutualistic value and task type.

7.3 Editing with breed- and task-type-conditional edit-power

The v2.1 edit-power formula is extended:

$$\text{edit-power} = \text{skill} \cdot (0.5 + 0.5 \cdot \text{motivation}) \cdot (1 + \text{topic-match} \cdot 0.50 + \text{explicit-bonus}) \cdot J$$

where $J = 1$ for humans on any patch, and for AI agents:

$$J = 1 - \text{judgment-weight} \cdot \text{judgment-penalty-strength}$$

with judgment-penalty-strength = 1.0. On "prediction" patches (judgment-weight = 0) or for any human editor, $J = 1$ always — exact v3.0/v2.2 equivalence. All other patch-update rules (quality gain, completeness gain, pheromone deposit, resource-value gain, maintenance relief, SMM/co-editing update) are unchanged from v2.1, **except** that AI agents never grow their own shared-mental-model on any succession.

7.4 AI-trust update (humans only)

Fires only when a human edits a patch whose previous editor was an AI agent (captured via last-editor-ai?, already tracked for the v3.0 SMM logic). The AI's prior contribution is judged using the patch's quality value as it stood **before** the human's own edit (pre-edit-quality):

$$\text{is-error} = \text{percentile-rank}(\text{pre-edit-quality} \mid \text{touched patches}) \leq 25$$

That is, the AI's prior edit counts as an observed "error" if its quality ranks at or below the 25th percentile of the current quality distribution across all patches with at least one edit — a **population-relative**, scale-free threshold (not a fixed absolute value), so it self-calibrates to

whatever quality scale a given parameter set produces and needs no manual recalibration if edit-increment, quality-gain-factor, or run length change.

- **On error:** ai-trust = max(0, ai-trust − ai-trust-penalty) with ai-trust-penalty = 0.40; ai-consecutive-success resets to 0; ai-error-counter increments (diagnostic only).
- **On success:** ai-consecutive-success increments; once it reaches ai-trust-recovery-threshold = 3 consecutive successes, ai-trust = min(1, ai-trust + ai-trust-recovery) with ai-trust-recovery = 0.15, and the counter resets to 0.

This asymmetry (fast single-event erosion, slow multi-event recovery) operationalises the design premise that trust in AI is fragile. At ai-share = 0, no patch is ever last-edited by an AI agent, so this block never fires and ai-trust never leaves its initial value (0.5) — harmless and inert, exact v3.0/v2.2 equivalence.

7.5 Trust-modified benefit

For a human editing a patch whose previous editor was an AI agent, the local-benefit formula from v2.1 (§7.3) is modified:

$$B_{\text{modified}} = B \cdot (0.5 + 0.5 \cdot \text{ai-trust})$$

where B is the unmodified v2.1 local-benefit (0.40·resource-value + 0.20·quality + 0.30·topic-match). A fully distrustful human (ai-trust → 0) derives only half the usual benefit from an AI-touched artefact, even if its measured quality is objectively fine — perceived value is trust-gated. A fully trusting human (ai-trust → 1) derives the full, unmodified benefit. All other succession types (human-following-human, any AI-agent editor) use the unmodified v2.1/v2.2 formula.

7.6 Fixed internal parameters (new in v3.0/v3.1)

All v2.1 fixed parameters (§7.5 of the v2.1 ODD) are unchanged. New:

Parameter	Value	Meaning

ai-motivation-constant	1.0	Fixed motivation assigned to every AI agent
judgment-weight-min	0.4	Lower bound of random judgment-weight draw
judgment-weight-max	1.0	Upper bound of random judgment-weight draw
judgment-penalty-strength	1.0	Scales how much judgment-weight reduces AI edit-power
ai-trust-initial	0.5	Starting ai-trust for humans (agnostic prior)
ai-trust-error-percentile	25	Percentile cutoff for judging an AI edit an "error"
ai-trust-penalty	0.40	ai-trust drop on one observed error
ai-trust-recovery	0.15	ai-trust gain after a qualifying success streak
ai-trust-recovery-threshold	3	Consecutive successes required before recovery is applied

8. Simulation Experiments and Key Results

This section documents the two new experiments introduced for STiMUS-HAI,extending the seven v2.1 experiments (E1–E7, unaffected, see the v2.1 ODD).E11/E12: 500 ticks per run, NetLogo 7.

8.1 E11 — Human-AI team composition (ai-share × vision-radius × p-explicit × mutualism-gain-factor)

RQ. If a human team is mixed with a varying share of AI agents, what happens to collaboration quality, and does a "camouflage" effect (H-E11.3) hide a real drop in human-human coordination behind healthy-looking aggregate indicators?

Design. Full factorial: ai-share ∈ {0, 0.10, 0.25, 0.50} × vision-radius ∈ {2, 5} × p-explicit ∈ {0, 0.5, 1} × mutualism-gain-factor ∈ {0, 0.06, 0.15}; 50 repetitions; 3600 runs total.

Result — backward compatibility. At ai-share = 0: ESR-all = ESR-HH = ESR; human-edit-share = 100%; AI pheromone contributions = 0. Exact v2.2 equivalence confirmed.

Result — aggregate productivity indicators are nearly blind to ai-share.

Indicator	η^2(ai-share)
mean-quality	0.0025
mean-resource-value	0.0000067
task-completion-rate	0.00014
gini-edits	0.0007
coediting-density	0.0002

Result — the camouflage effect (H-E11.3).

ai-share	ESR-all	ESR-HH	gap
0.00	0.675	0.675	0
0.10	0.676	0.589	0.087
0.25	0.676	0.507	0.169
0.50	0.675	0.309	0.366

ai-share explains 79% of the variance in the gap ($\eta^2 = 0.789$) and nearly half the variance in ESR-HH itself ($\eta^2 = 0.491$), but almost none in ESR-all ($\eta^2 \approx 0.000009$). At ai-share = 0.5, the observed ESR-HH (0.309) falls further than the naive combinatorial floor of 0.169 (expected if the drop were purely due to fewer human-human pairs being available) would suggest it *should* fall to — but is still well above that floor, and the "excess drop" relative to the naive floor is consistently negative (−0.042 to −0.140): the actual fall is less steep than the naive combinatorial model predicts, though still substantial relative to the baseline. Part of the fall is arithmetic (fewer human pairs are physically possible); a meaningful share reflects real structural change in how edit chains form.

Result — human SMM erosion (H-E11.1, not fully supported).

ai-share	human-smm
0.00	0.904
0.10	0.881
0.25	0.856

0.50	0.742

ai-share explains 20% of human-smm variance ($\eta^2 = 0.200$); even a modest 10% AI share already lowers human-smm measurably. mutualism-gain-factor does not mitigate this ($\eta^2 < 0.001$ in cross-terms with ai-share).

Result — visibility amplifies, not dampens, the camouflage effect.

ai-share	gap, vision=2	gap, vision=5
0.10	0.069	0.105
0.25	0.131	0.207
0.50	0.284	0.449

Wider vision-radius increases exposure to AI-authored pheromone trails, lengthening mixed human/AI succession chains and widening the gap — a counter-intuitive result: tools that broaden visibility of system activity do not protect against, and can worsen, this camouflage effect.

8.2 E12 — Task-type regime (ai-share × judgment-share × vision-radius)

RQ. Does the *type* of work AI performs — routine/prediction versus judgment-requiring — change how AI presence affects team outcomes, beyond simply *how much* AI is present?

Design. A pilot (160 runs, 10 reps) followed by the main factorial: ai-share ∈ {0, 0.15, 0.30, 0.45, 0.60, 0.75, 0.90} × judgment-share ∈ {0, 0.15, 0.30, 0.45, 0.60, 0.75, 0.90} × vision-radius ∈ {2, 5, 8}; 100 repetitions; 14,700 runs total.

Result — backward compatibility. At ai-share = 0: ai_edits, ai_pheromone_share, ai_errors all = 0; human_edit_share = 1; ESR-all = ESR-HH; ai_trust = 0.5 with zero variance. At judgment-share = 0: judgment_edit_share = 0, prediction_edit_share = 1, human_smm_judgment = 0. Both confirmed across the full 14,700-run grid.

Result — dominant main effects (η^2).

Dependent variable	dominant driver	η^2

ESR_all, coediting, task_comp	vision-radius	0.93, 0.94, 0.81
ESR_HH, esr_gap, human_smm	ai-share	0.79, 0.76, 0.66
judgment_sh_ed (judgment-edit share)	judgment-share	0.93
quality, resource, ai_trust	ai-share	0.50–0.53

Result — H-E12.1 (judgment-share steepens the AI dilution effect onESR-HH) — rejected. Interaction ai-share × judgment-share on ESR_HH: $\eta^2 = 0.00015$. ESR-HH falls with ai-share almost identically regardless of judgment-share (e.g. at ai-share = 0.9: ESR_HH = 0.0353 at judgment-share = 0 vs. 0.0311 at judgment-share = 0.9).

Result — H-E12.2 (judgment-share accelerates ai-trust erosion) —rejected in its direct causal form; confirmed as a mediated/measurementartifact. The direct interaction ai-share × judgment-share on ai_trust is modest ($\eta^2 = 0.035$), smaller than the interaction on quality ($\eta^2 = 0.136$) — judgment-share primarily depresses ai_trust **through** its effect on the population's quality distribution (the percentile-based error threshold, §7.4), not through an independent trust-erosion channel. This was identified and documented in the model code itself as a revision (see the REVISION NOTE at the ai-trust-quality-threshold reporter): an earlier absolute-threshold design produced noise-dominated ai-trust estimates; the population-relative percentile design (used in this release) resolves it.

Result — H-E12.3 (AI's effect on quality is regime-conditional) —strongly supported.

Regime	ai-share = 0	ai-share = 0.9	relative change
Judgment-heavy	0.194	0.116	−40%
Prediction-heavy	0.193	0.178	−8%

Task-completion and human-smm follow the same pattern (judgment-heavy: task_comp 0.812 → 0.576, human_smm 0.939 → 0.252; prediction-heavy: task_comp 0.811 → 0.784, human_smm 0.939 → 0.250 — note human_smm

collapses similarly in both regimes at very high ai-share, but quality and task_comp do not). The same nominal AI share is far more costly to work quality when most of the artefact base requires situated judgment than when it is mostly routine.

8.3 Summary of new hypotheses

Hypothesis	Experiment	Outcome
H-E11.1 Moderate AI ≠ harmless to human cognition	E11	Not fully supported — even 10% AI lowers human-smm
H-E11.2 AI offloads coordination without harming outcomes	E11	Partially supported — aggregate productivity unaffected
H-E11.3 Aggregate ESR camouflages human-human erosion	E11	Strongly supported (η^2(ai-share→esr_gap) = 0.79)
H-E12.1 judgment-share steepens AI dilution of ESR-HH	E12	Rejected (interaction $\eta^2 \approx 0.00015$)
H-E12.2 judgment-share accelerates AI-trust erosion (direct)	E12	Rejected direct form; mediated by quality
H-E12.3 AI's quality cost is regime-conditional	E12	Strongly supported (−40% vs. −8%)

9. Interpretation for Pedagogical and Organisational Design

1. **Aggregate KPIs are not sufficient to monitor mixed human-AI teams.** E11 shows that quality, completion rate, and coediting density can look completely healthy while human-to-human coordination (ESR-HH, human-smm) quietly deteriorates. Design and evaluation of collaborative platforms with AI participants needs author-pair-stratified metrics, not just team-level aggregates.
2. **The cost of AI presence depends on what AI is asked to do, not only howmuch of it there is.** E12's central finding: the same ai-share can be nearly costless (prediction-heavy regime) or substantially damaging to quality (judgment-heavy regime). This reframes "how much AI should this team have" as "what proportion of *this team's work* genuinely requires situated judgment".

3. **Visibility tools are not a universal remedy.** Widening `vision-radius` — the analogue of more transparent activity feeds or dashboards — worsens rather than mitigates the ESR-all/ESR-HH camouflage effect at high AI share (E11 §8.1).
4. **Trust modelling should be population-relative, not absolute.** The model's own revision history (§7.4, the percentile-based error threshold) is itself a design lesson: fixed absolute quality thresholds for "AI made an error" are fragile to arbitrary parameter choices and produce noise-dominated trust estimates; a population-relative standard is more robust and more conceptually faithful to how humans actually judge "typical" versus "substandard" work.

10. Limitations and Future Extensions

All v2.1 limitations (§10 of the v2.1 ODD) apply unchanged where relevant. New for v3.0/v3.1:

1. **ai-agents is a single homogeneous breed.** No sub-typing by AI reliability, specialisation, or model family. The v3.1 design document argues this split is only warranted once a factorial experiment shows patterns that a single `ai-share` parameter cannot explain — E11 and E12 found no such signal.
2. **No Q-learning or other adaptive action-selection for AI agents.** AI agents follow the same fixed stigmergic/explicit rule as humans; a Manseau-style Q-learning layer was considered and explicitly deferred (see the Manseau integration design notes) to preserve comparability with v2.2/v3.0.
3. **judgment-weight is drawn once at setup and does not evolve.** A more realistic model might let task difficulty or AI capability change over a run.
4. **The "expert paradox" (do higher-skill humans lose AI trust faster?) isnot directly modelled** — `ai-trust` is tracked per-human and exportable,

but correlating it with `skill` is left to R-side post-processing on the exported edge log, not computed as a NetLogo reporter.

5. **H-E12.2's rejection is itself informative but not exhaustivelydiagnosed.** The mediating role of quality is documented, but a full causal decomposition (e.g. formal mediation analysis) was not performed in this release.

References

Grimm, V., Berger, U., Bastiansen, F., et al. (2006). A standard protocol for describing individual-based and agent-based models. *Ecological Modelling*, 198(1–2), 115–126.

Grimm, V., Berger, U., DeAngelis, D. L., Polhill, J. G., Giske, J., & Railsback, S. F. (2010). The ODD protocol: A review and first update. *Ecological Modelling*, 221(23), 2760–2768.

Internal design document: "STiMUS_Manseau_integration.md" — analysis of a Manseau-style Q-learning human-AI collaboration model, informing the ai-trust mechanic design (§3.1) and the explicit decision to defer Q-learning action selection.